

Three Planets, No Formula: What Computers Can Tell Us About the Three-Body Problem

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1 What is the three-body problem?

Imagine three stars floating in space, each one pulling on the other two with gravity. Two stars orbiting each other is easy: you can write down a formula for the path they trace out, and that formula works forever. Three stars is a different story. Isaac Newton wrote down the rule for how gravity pulls two things together over 300 years ago, and it works perfectly well for three stars too, in the sense that you can always calculate what happens *next*. The trouble is that nobody has ever found one single formula that predicts what three gravitating bodies will do far into the future, the way we can for two. In 1890, the mathematician Henri Poincaré proved that, in general, no such formula can exist.

Two special cases were already known long before Poincaré's proof. In 1767, Leonhard Euler found an exact solution in which all three bodies stay lined up on a single rotating straight line. In 1772, Joseph-Louis Lagrange found a second exact solution in which the three bodies stay at the corners of a rotating equilateral triangle, whatever their individual masses. Both are genuine, exact three-body solutions – but Euler's lined-up version is always unstable, like balancing a pencil on its tip, while Lagrange's triangle is only stable if one of the three bodies is much heavier than the other two. What Poincaré's proof shows is that beyond these two special families, no single formula can cover every possible three-body arrangement.

That does not mean three-body motion is unpredictable in every sense. A computer can simulate it step by step, as far forward or backward in time as you like, and check its own work along the way. This project used computer simulations to ask two specific questions about three-body motion, and to see how far a computer alone – without copying a known answer from a textbook – can get.

2 Two big questions

Question 1. Some three-body paths are special: they loop back on themselves exactly, tracing the same shape over and over, forever, like a racetrack. These are called *periodic orbits*. If we do not start from a racetrack shape we already know about, can a computer search find a brand-new one from scratch?

Question 2. Not every useful three-body system has to loop back on itself. Sometimes all we want is a system that will stay together – no crashes, nothing flying off into space – for some amount of time we choose in advance. If we pick a length of time, can we build, or find, a three-body system that we can show will last at least that long?

These two questions sound similar, but as this project found, they have very different answers.

3 Why three bodies are so hard to predict

With two bodies that are gravitationally bound to each other – moving slowly enough that neither can break free of the other’s pull – gravity settles into one clean repeating shape: a circle or an ellipse, retraced forever. (If the two bodies are moving too fast to stay bound, the path is an open curve, a parabola or hyperbola, and they separate for good instead of repeating – but bound, repeating orbits are the common case, and the one this project is concerned with.) Add a third body, and the extra tug of gravity usually breaks that clean, repeating pattern. The system becomes *chaotic*: a tiny change in where you place one of the three bodies at the start – a change too small to ever measure in real life – can lead to a completely different outcome after enough time. Two nearly identical three-body systems can end up looking nothing alike a little while later. This is the same “butterfly effect” idea used to describe why weather is hard to predict more than a week or two out.

Because of this, almost no starting arrangement of three bodies leads to a tidy repeating loop. Most either fly apart, or two of the bodies get close enough to collide. Loop-forming arrangements are more like winning numbers in a lottery: they exist, but you have to search for them, and most tickets lose.

4 Question 1: searching for a brand-new repeating orbit

We tried four different ways of searching for a new repeating three-body orbit, and in every case, we made sure the search itself never used a known orbit as a shortcut or a hint.

What we tried. We started bodies lined up in a row, moving at different angles; we started them at rest in a triangle shape and let gravity pull them together; we started them lined up and moving straight away from the line; and finally we spread the search out over a whole two-dimensional map of starting shapes, throwing out any attempt that got dangerously close to a crash before even trying to refine it.

What happened. In every case except the last, almost every starting point we tried ended the same way: two or three of the bodies crashed into each other, often within just a couple of time steps. In the last, broader search, more than half of our attempts avoided crashing – but even so, none of them turned into an exact, repeating loop.

Why. A spinning top stays upright partly because of its spin. Something similar happens with orbits: when a group of bodies has some overall “spin” (physicists call it angular momentum), that spin acts like a cushion that tends to keep the bodies apart. Every starting arrangement we tried in this project had *zero* overall spin. Without that cushion, there is nothing to stop the bodies from drifting toward each other and colliding, which is exactly what we saw happen over and over.

One well-known repeating three-body orbit, called the **figure-eight orbit**, also has zero overall spin, and yet it does not collide – the three bodies trace out a graceful figure-eight shape forever, chasing each other around the same curve. It was first found numerically by Moore in 1993 [Moore, 1993], and its existence was later proved rigorously, not just numerically, by Chenciner and Montgomery in 2000 [Chenciner and Montgomery, 2000]. We used it to check that our simulation was working correctly (Figure 1), by measuring how precisely it retraces its own path – it comes back to where it started to within about one part in a trillion. Finding an orbit like this from scratch, though, is like finding one four-leaf clover in an enormous field: it can be done, but researchers who have actually found new examples – such as Šuvakov and Dmitrašinović’s 2013 search, which turned up 13 new families [Šuvakov and Dmitrašinović, 2013], and Li and Liao’s later, much larger catalogs [Li and Liao, 2017, Li et al., 2018] – needed far more computing time than we used here,

plus a clever way of first sorting candidates by the general shape of the path they trace, before checking any of them in detail.

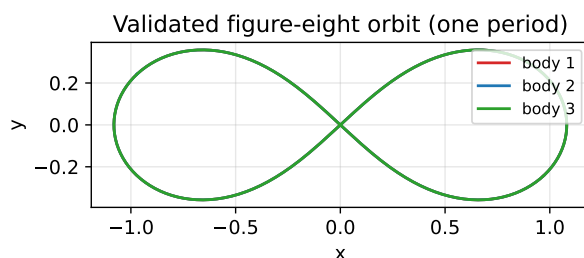


Figure 1: The figure-eight orbit: three equal bodies chasing each other around one shared path, forever. [\[watch: watch it move\]](#).

Bottom line for Question 1: yes, in principle, a computer search that starts from nothing can find brand-new repeating orbits – other researchers have done it. But it takes a lot more computing power, or a smarter search strategy, than a simple scan. Our own simple scan did not turn up a new one, and we can point to exactly why: without any spin to keep the bodies apart, most of the space we searched was really the space of ways to crash, not the space of ways to orbit.

5 Question 2: building a system that lasts

Here the news is much better – and we found two completely different ways to answer this question.

Method 1: keep a tight pair, and put the third body far away

Picture two dancers holding hands and spinning quickly in a small circle, while a third dancer slowly circles around the outside, far from the pair. As long as the outside dancer stays far enough away, the inner pair barely notices they are there, and keeps spinning steadily.

The same idea works for gravity. We built several three-body systems with a tight inner pair and a third body circling far outside, and changed only one thing between them: how far away the outside body was, compared to the size of the inner pair’s own little orbit.

- When the outside body was **far away** (about 6 times the inner pair’s own orbit size), the system was still running smoothly after **500 complete loops** of the inner pair – we simply let the simulation run that long and watched nothing go wrong.

- When the outside body was placed **too close** (under 2 times the inner pair’s orbit size), the system fell apart after only about **18 loops** – the outside body’s pull was strong enough to yank the inner pair out of shape.

There turns out to be a fairly simple rule of thumb, worked out by other scientists [Mardling and Aarseth, 2001] and confirmed again in this project, for exactly how far away is “far enough” to guarantee the tight pair survives. That means we did not have to search or guess: we could simply calculate the right spacing in advance, build the system that way, and it worked exactly as predicted (Figure 2).

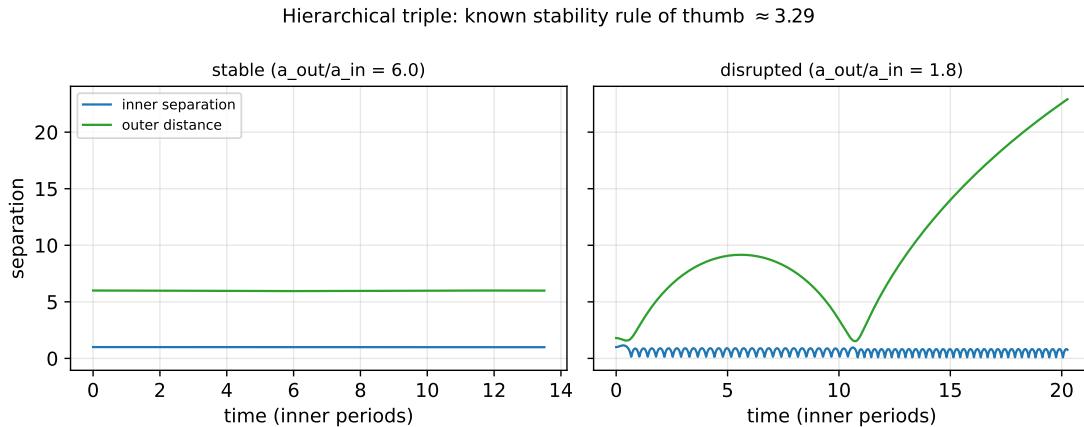


Figure 2: Left: outside body far away – everything stays tightly bound. Right: outside body too close – the inner pair is pulled apart. [\[watch: watch the difference\]](#).

Method 2: try many random setups, and see how many survive

Not every three-body system has a neat “tight pair plus outsider” shape. So we also asked a more general question: if you set up 200 completely random three-body systems – random shapes, random directions of motion – what fraction of them are still intact after a given amount of time?

The answer, out of our 200 random trials:

- About **45 out of 100** ended in a collision.
- About **41 out of 100** ended with one body flying away for good.
- About **14 out of 100** were still together, undisturbed, at the very end of the time we tested.

The longer you want a random system to survive, the smaller that last group gets, but it never quite reaches zero within the time we tested. That means that if you want a system guaranteed (or even just likely) to last a certain amount of time, and you are willing to try more than one random setup, you can simply generate a batch, run them forward, and pick one of the survivors (Figure 3).

Bottom line for Question 2: yes, and in two different ways. If your system has a natural “tight pair plus one outsider” shape, a known rule tells you exactly how to space it so it lasts as long as you want, with no trial and error. If it does not have that shape, you can still get a good, verified survivor by generating several random attempts and checking which ones last – you just cannot guarantee any single one in advance the way you can with the tight-pair method.

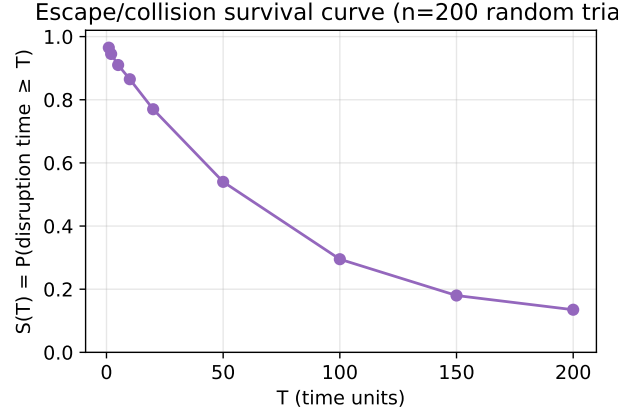


Figure 3: Out of 200 random three-body setups, this shows what fraction were still intact after a given amount of time. [\[watch: watch some examples\]](#).

6 What we learned

A computer cannot hand you one formula that predicts three-body motion forever, the way it can for two bodies. But it turns out that “can we predict it” is really two separate questions, and they have different answers. Finding a brand-new repeating three-body orbit from nothing is genuinely hard – our own attempts all failed, and we could see exactly why: without any spin to keep the bodies apart, most starting points simply lead to a collision instead of a graceful loop. But building a three-body system that survives for as long as you want is much more approachable, whether by using a known spacing rule for a tight pair plus an outsider, or by testing a batch of random setups and keeping the ones that hold together.

The three short videos linked throughout this document show all of this in motion: the figure-eight orbit tracing its endless path, one three-body system staying calmly together while another one is torn apart, and a batch of random three-body “experiments” playing out to see which ones survive.

All of the code is available at <https://github.com/adityabhattacharjee/three-body-problem>.

References

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